



Modelling Financial Markets

Seminar 14: Deep Learning and Time-Series Foundation Models — Hands-on



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Acronyms used in this material

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- CI: Confidence Interval
- DM: Diebold–Mariano (test)
- ES: Expected Shortfall
- FHS: Filtered Historical Simulation
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- HAR: Heterogeneous AutoRegressive (model)
- HS: Historical Simulation
- LSTM: Long Short-Term Memory (network)
- MCS: Model Confidence Set
- QLIKE: Quasi-LIKElihood (loss function)
- RNN: Recurrent Neural Network
- RV: Realised Variance

Outline

Introduction

Key Formulas

Part A: Derivations with Numbers

Part B: Inference on the Forecasts

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Count the parameters of an LSTM and quantify how fast its memory fades
- ▣ Tokenise a series as Chronos does and read VaR and ES from a quantile grid
- ▣ Test whether a deep or foundation model beats a strong benchmark (DM, MCS, Kupiec, FZ0)
- ▣ Detect seed effects and look-ahead leakage

- ▣ Background: Lecture 14 and Lecture 8 (backtesting); textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations with numbers (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Always name the benchmark: zero for returns, log-HAR for volatility, FHS for tail risk
- A model that is "not significantly worse" is not "as good": ask how much power the test had

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (LSTM parameters), A3 (memory of the forget gate), A5 (tokenisation), A7 (VaR and ES from a quantile grid)
- Part B: B1 (returns vs zero, S&P 500), B3 (realised volatility vs log-HAR), B5 (VaR 1% and ES 2.5%, S&P 500), B7 (seeds)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5), A8 (model: A7)
- Part B: B2 (model: B1), B4 (model: B3), B6 (model: B5), B8 (model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (a foundation-model VaR backtest)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter14_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `torch`, `chronos-forecasting`, `timesfm`, `arch`, `pandas`, `scipy`
- Series: S&P 500, Bitcoin, BET (daily closes); SPY 5-minute bars for realised variance; up to 18 September 2026
- Forecasts: the one-day-ahead forecasts of Lecture 14 (context of 512 days for the foundation models; windows of 500 or 1,000 days for the classical models)
- Returns r_t in %, losses $L_t = -r_t$; $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$

Key Formulas (1/2): Networks and Tokens

- LSTM (Hochreiter & Schmidhuber (1997)): $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, $h_t = o_t \odot \tanh(c_t)$; gates $\sigma(Wx_t + Uh_{t-1} + b) \in (0, 1)$
- Parameters of one layer: $4(hd + h^2 + h)$, or $4(hd + h^2 + 2h)$ with two bias vectors (PyTorch)
- Memory along the cell state: $\partial c_T / \partial c_{T-k} = \prod_j f_{T-j}$; half-life $k_{1/2} = \ln 0.5 / \ln f$
- Attention (Vaswani et al. (2017)): $\text{softmax}(QK^\top / \sqrt{d_k}) V$
- Chronos tokens (Ansari et al. (2024)): $s = \frac{1}{\epsilon} \sum |x_t|$, $z_t = x_t / s$, bin index $j = \text{round}((z_t + 15) / \Delta)$, $\Delta = 30/4092$

Key Formulas (2/2): Evaluation

- Out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ (benchmark: zero forecast)
- Diebold–Mariano (Diebold & Mariano (1995)): $DM = \bar{d} / \sqrt{\widehat{LRV} / T}$, $d_t = L_t^A - L_t^B$, Newey–West long-run variance; negative: A better
- QLIKE (Patton (2011)): $RV_t / \hat{h}_t - \log(RV_t / \hat{h}_t) - 1$; HAR (Corsi (2009)): RV on its daily, 5-day and 22-day averages
- Kupiec (Kupiec (1995)): $LR_{uc} = -2[(T - x) \ln(1 - \alpha) + x \ln \alpha - (T - x) \ln(1 - \hat{\pi}) - x \ln \hat{\pi}] \sim \chi_1^2$
- FZ0 (Patton, Ziegel & Chen (2019)): $\frac{1}{\alpha \text{ES}} 1\{L > \text{VaR}\}(L - \text{VaR}) + \frac{\text{VaR}}{\text{ES}} + \log \text{ES} - 1$; MCS (Hansen, Lunde & Nason (2011)) at 90%

A1. Counting the Parameters of an LSTM [Solved]

Task

- One LSTM layer, input size $d = 1$, hidden size $h = 16$, linear output layer
- Count the parameters with one bias per block, then as PyTorch does (two bias vectors)
- Compare with a training sample of 4,000 daily returns

Solution

- Per block: $hd + h^2 + h = 16 + 256 + 16 = 288$; four blocks: 1 152
- PyTorch: $4(16 + 256 + 32) = 1\,216$; output layer $h + 1 = 17$; total 1 233
- About 3 observations per parameter: early stopping and weight decay are not optional

A2. Two Stacked Layers and Five Inputs [Proposed]

Solution discussed in class

Task

- ▣ Inputs: 5 features per day; two LSTM layers with $h = 32$; linear output
- ▣ Count all parameters (PyTorch convention); the second layer receives $d = 32$
- ▣ How many years of daily data give 3 observations per parameter?
- ▣ Model: A1

A3. How Fast Does Memory Fade? [Solved]

Task

- Simple RNN, scalar case: the gradient after k steps is at most $|w|^k$; take $|w| = 0.9$
- LSTM with a constant forget gate $f = \sigma(b)$ for $b = 0$ and $b = 3$
- Compute what is left after 20 and 50 steps, and the half-life

Solution

- RNN: $0.9^{20} = 0.122$, $0.9^{50} = 0.0052$
- $b = 0$: $f = 0.5$, $0.5^{50} \approx 8.9 \times 10^{-16}$
- $b = 3$: $f = 0.9526$, $f^{50} = 0.088$; half-life $\ln 0.5 / \ln f = 14.3$ steps
- The value for $b = 3$ matches the measured gradient at lag 50 in Lecture 14 (about 8%)

A4. Half-Lives of the Forget Gate [Proposed]

Solution discussed in class

Task

- ▣ Half-life for $b = 1$ and for $f = 0.99$
- ▣ Which bias b gives $f = 0.99$?
- ▣ With $b = 5$, what fraction survives one trading year (250 days)?
- ▣ Model: A3

A5. Chronos Tokenisation of a Four-Day Context [Solved]

Task

- Context (returns in %): $x = (0.8, -1.2, 0.4, 2.0)$
- Mean scaling, then 4,093 equally spaced bin centres in $[-15, 15]$
- Compute s , the scaled values and the bin indices

Solution

- $s = (0.8 + 1.2 + 0.4 + 2.0)/4 = 1.10$
- $z = x/s = (+0.727; -1.091; +0.364; +1.818)$
- $\Delta = 30/4092 = 0.00733$; $j = \text{round}((z + 15)/\Delta) = (2145, 1897, 2096, 2294)$
- The centre of the vocabulary is index 2046 ($z = 0$): all four days sit close to it

A6. A Crash Day in the Context [Proposed]

Solution discussed in class

Task

- ▣ Replace the last value by -9.0 : $x = (0.8, -1.2, 0.4, -9.0)$
- ▣ Recompute s , z and the bin indices
- ▣ How does the token of the 0.4% return change? What does the model "see" after a crash?
- ▣ Model: A5

A7. VaR 1% and ES 2.5% from a Quantile Grid [Solved]

Task

- Last Chronos-2 forecast for the S&P 500 (day after 18.09.2026): $q_{0.01} = -2.704\%$, $q_{0.05} = -1.400\%$
- Compute VaR 1%, VaR 2.5% (linear interpolation) and ES 2.5%, holding $q_u = q_{0.01}$ for $u < 0.01$

Solution

- $\text{VaR } 1\% = -q_{0.01} = 2.704\%$
- $q_{0.025} = q_{0.01} + \frac{0.015}{0.04}(q_{0.05} - q_{0.01}) = -2.215\%$
 $\text{VaR } 2.5\% = 2.215\%$
- $\text{ES } 2.5\%$
 $= -\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.557\%$
- ES/VaR at 2.5%: 1.15; a lower bound, since the tail below 1% is cut

A8. A "VaR 1%" from Deciles [Proposed]

Solution discussed in class

Task

- ▣ Chronos-Bolt, same day: $q_{0.1} = -0.691\%$, $q_{0.5} = 0.071\%$; no level below 10%
- ▣ (a) the value returned when VaR 1% is requested; (b) a Normal extrapolation: $\hat{\sigma} = (q_{0.5} - q_{0.1})/z_{0.9}$, VaR 1% and ES 2.5%
- ▣ Compare with Chronos-2 in A7
- ▣ Model: A7

B1. Can Anything Beat Zero for the S&P 500? [Solved]

Task

- One-day-ahead return forecasts of six models, S&P 500, 2 693 days from 2016
- Out-of-sample R^2 with a 95% block-bootstrap CI (blocks of 20 days)
- DM test on squared errors against the zero forecast; share of correct signs

Interpretation

- Is a positive R^2 of about 1% statistically meaningful here?
- Why is 55% correct signs not evidence of skill?

B1. Results

Model	R^2 (%)	95% CI	DM [p]	Correct sign (%)
Historical mean	+0.15	[-0.02; +0.40]	-1.67 [0.095]	54.6
AR(1)	+0.94	[-2.65; +4.40]	-0.47 [0.639]	54.1
LSTM	+0.82	[+0.13; +1.43]	-1.57 [0.116]	51.2
Chronos-2	-0.26	[-0.89; +0.38]	+0.81 [0.418]	54.7
Chronos-Bolt	-1.57	[-2.73; -0.24]	+1.76 [0.078]	54.6
TimesFM-2.5	-3.35	[-4.62; -1.95]	+3.23 [0.001]	53.6

- No CI excludes zero on the positive side except the LSTM ([+0.13; +1.43]), yet its DM p is 0.116: the evidence is weak
- TimesFM-2.5 is significantly worse than zero; AR(1) has a huge CI, driven by 2020
- The S&P 500 rose on 54.6% of days: "always up" already gets that share of signs right

B2. Bitcoin and BET [Proposed]

Solution discussed in class

Task

- Repeat B1 for Bitcoin (2818 days from 2019) and the BET (2675 days from 2016)
- Model: B1

Interpretation

- Which series has predictable daily returns, and which model finds it?
- Relate the answer to the efficiency results of Chapter 2

B3. Realised Volatility: Better than log-HAR? [Solved]

Task

- ▣ SPY realised variance, 979 one-day-ahead forecasts; eight models
- ▣ Mean QLIKE, DM against log-HAR (Newey–West), 90% MCS
- ▣ Repeat on the 218 days after 3 November 2025

Interpretation

- ▣ What does membership of the MCS tell you, and what does it not tell you?
- ▣ Why does the post-release window matter for foundation models only?

B3. Results

Model	QLIKE	DM vs log-HAR [p]	MCS p	QLIKE (post)	DM (post)
Random walk	0.326	+4.53 [<0.001]	0.00	0.384	+2.81
HAR	0.247	+1.30 [0.194]	0.26	0.270	+1.40
log-HAR	0.234	–	0.95	0.238	–
GARCH-t	0.283	+3.43 [<0.001]	0.03	0.295	+2.34
LSTM	0.290	+1.87 [0.062]	0.19	0.231	-0.61
Chronos-2	0.231	-0.62 [0.537]	1.00	0.242	+0.58
Chronos-Bolt	0.232	-0.18 [0.857]	0.95	0.236	-0.39
TimesFM-2.5	0.233	-0.08 [0.938]	0.95	0.244	+0.60

- Chronos-2 has the lowest QLIKE but is not significantly better than log-HAR (DM -0.62, $p = 0.537$)
- MCS: random walk and GARCH-t are eliminated; the MCS says "not distinguishable", not "equally good"
- After 3.11.2025 the LSTM is best (0.231); no DM test is significant against log-HAR except for the random walk and GARCH-t

B4. Median or Mean? And How Much Context? [Proposed]

Solution discussed in class

Task

- ▣ For Chronos-2 and TimesFM-2.5, compare $\exp(q_{0.5})$ with $\int_0^1 \exp(q_u) du$ on the grid
- ▣ QLIKE of Chronos-2 with contexts of 32, 64, 128, 256 and 512 days
- ▣ Model: B3

Interpretation

- ▣ Why is the median biased as a variance forecast?
- ▣ What does the context effect say about the memory of volatility?

B5. Backtesting the Tail Forecasts on the S&P 500 [Solved]

Task

- ▣ Six models: HS, GARCH-t, FHS, raw Chronos-2 quantiles, FHS with Chronos-2 or TimesFM volatility
- ▣ VaR 1%: breaches, Kupiec, Christoffersen independence
- ▣ (VaR 2.5%, ES 2.5%): mean FZ0, DM against FHS

Interpretation

- ▣ Why do the raw Chronos-2 quantiles fail while the hybrid passes the coverage test?
- ▣ Passing coverage, does the hybrid match FHS?

B5. Results (S&P 500, 2 693 days)

Model	Breaches	Rate (%)	p Kupiec	p indep.	FZ0	DM vs FHS
HS	37	1.37	0.065	0.001	1.309	+2.45
GARCH-t	44	1.63	0.002	0.205	1.026	+1.34
FHS	30	1.11	0.559	0.045	0.999	–
Chronos-2 quantiles	45	1.67	0.001	0.223	1.147	+3.16
FHS + Chronos-2 vol.	25	0.93	0.705	0.021	1.084	+2.65
FHS + TimesFM vol.	27	1.00	0.989	0.029	1.059	+1.89

- Raw Chronos-2: 1.67% breaches, Kupiec $p = 0.001$: its quantiles are too narrow
- Hybrid (Chronos-2): 0.93%, $p = 0.705$: the tail comes from the data, through FHS
- But FZ0 is significantly worse than FHS (DM +2.65): correct coverage with too much capital

B6. Bitcoin, and VaR 10% for All Three Models [Proposed]

Solution discussed in class

Task

- Repeat B5 for Bitcoin (2818 days)
- VaR 10%: breach rates and Kupiec tests of Chronos-2, Chronos-Bolt and TimesFM-2.5 quantiles against HS and FHS, in the three markets
- Model: B5

Interpretation

- Is the narrowness of foundation-model quantiles specific to the 1% level?

B7. The Seed Lottery [Solved]

Task

- ▣ Five LSTMs for returns, identical except for the random seed
- ▣ Out-of-sample R^2 of each seed and of their average, three markets

Interpretation

- ▣ What would you conclude from a single run on Bitcoin?
- ▣ Why is averaging seeds a form of regularisation?

B7. Results

R^2 (%)	Seed 0	1	2	3	4	Average
S&P 500	+0.53	+0.35	+0.48	+0.55	+1.40	+0.82
Bitcoin	-0.08	+0.02	-1.47	-0.49	+0.21	-0.08
BET	+1.10	+1.03	+1.15	+1.25	+1.14	+1.18

- Bitcoin: seeds range from -1.47% to +0.21%: a single run could "show" skill or its absence
- BET: all five seeds agree; averaging reduces variance without adding bias, like bagging

B8. Leakage Through a Look-Ahead Regressor [Proposed]

Solution discussed in class

Task

- log-HAR for SPY, but with the 5- and 22-day averages computed up to and including the forecast day t
- Compare its QLIKE with the correct log-HAR and with Chronos-2; DM test
- Model: B3

Interpretation

- How large is the gain from the look-ahead? Could any honest model get it?
- Where can the same error hide in a deep-learning pipeline?

C1. Can a Zero-Shot Foundation Model Replace GARCH for the BET? [Proposed]

Solution discussed in class

Task

- ▣ A risk manager proposes Chronos-2 for the daily VaR 1% and ES 2.5% of a BET position, instead of GARCH-based models
- ▣ Compare HS, GARCH-t, FHS, raw Chronos-2 quantiles and FHS with foundation-model volatility (2 675 days from 2016)
- ▣ Use coverage, independence, FZ0, DM, MCS, Basel zones on 250-day blocks and the post-release window

Interpretation

- ▣ Write a one-page recommendation: accept, reject, or accept as a challenger model?
- ▣ Project extension: fine-tune a foundation model on Romanian stocks and re-run the test

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Backtest zero-shot Chronos-2 one-day-ahead VaR 1% for the S&P 500 on the last 1500 days, with a 512-day context”
- The assistant answers with this code (spx: daily closes, imports omitted):

```
pipe = BaseChronosPipeline.from_pretrained('amazon/chronos-2')
r = 100 * np.log(spx).diff().dropna().to_numpy()      # daily returns, %
test = np.arange(len(r) - 1500, len(r))
q01 = np.empty(len(test))
for i, t in enumerate(test):                          # forecast of r[t]
    ctx = torch.tensor(r[t - 511:t + 1], dtype=torch.float32) # 512 days
    q, _ = pipe.predict_quantiles([ctx[None, :]], prediction_length=1,
                                  quantile_levels=[0.01])
    q01[i] = q[0][0, 0, 0].item()
x = int((r[test] < q01).sum())                          # breaches
print(f"VaR1%={q01.mean():.2f}%,{breaches}{x}")
```

- “Output: VaR 1% = −2.41%, breaches = 5, i.e. 0.33% against 15 expected: the model is conservative, so it passes the Kupiec test easily”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains **three** errors, in the code and in the conclusion; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: an index check, the VaR convention of the lecture, the test statistic
 - ▶ give the corrected line or conclusion
- Compute the Kupiec statistic and its p -value for 5 breaches in 1 500 days
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Which element of $\mathbf{r}$ should be the last one in the context when forecasting $\mathbf{r}[t]$?
- Is “too few breaches” good news for a risk manager? For a regulator?
- The weights of Chronos-2 were published in autumn 2025: is the last 1 500 days a clean test for it?

Take-Home Assignment (Team Project)

- ▣ Choose one asset not covered in class (a BVB stock, a crypto-asset, an exchange rate or a commodity)
- ▣ Forecast its daily volatility or VaR 1% with one foundation model, zero-shot, and with a GARCH benchmark
 - ▶ Rolling origin, and the context window must end the day before the forecast
 - ▶ Report the period after the model's release date separately
- ▣ Compare the models with a DM test and, for VaR, with the coverage and independence tests
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

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